Example 6.14. In Example 6.11, we abstracted the state of the process by two variables p and c. The next step is to model the process of cutting the chocolate bar. When we make a single cut of a piece, the number of pieces (p) and the number of cuts (c) both increase by 1. We can model it by an assignment statement.

p, c := p + 1, c+1

which is read as p and c "become" p + 1 and c + 1, respectively.

Points to Remember:

- A programming language provides basic statements and a notation for composing compound statements.
- An algorithm is a step-by-step sequence of statements to solve a problem.
- As an algorithm is executed, a process evolves which solves the problem.
- Algorithmic problem solving involves construction of algorithms as well as proving properties of algorithms.
- The specification of an algorithm consists of the name of the algorithm (together with its inputs), the input property, and the desired input-output relation.
- Specification of an algorithm is a contract between the designer and users of the algorithm.
- Abstraction is the process of hiding or ignoring the details irrelevant to the task so as to model a problem only by its essential features.
- Specification abstracts a problem by the essential variables of the problem.
- The values of the variables in an algorithm define the state of the process.
- Assignment statement changes the values of variables, and hence the state.

Evaluation

IFEDO

SECTION – A

Choose the correct answer

1. Which of the following activities is algorithmic in nature?

   (a) Assemble a bicycle. (b) Describe a bicycle.
   (c) Label the parts of a bicycle. (d) Explain how a bicycle works.

2. Which of the following activities is not algorithmic in nature?

   (a) Multiply two numbers. (b) Draw a kolam.
   (c) Walk in the park. (d) Swaping of two numbers.

3. Omitting details inessential to the task and representing only the essential features of the task is known as

   (a) specification (b) abstraction (c) composition (d) decomposition

4. Stating the input property and the input-output relation a problem is known

   (a) specification (b) statement (c) algorithm (d) definition

11th Std Computer Science_EM Chapter 6.indd 85 06-01-2023 15:06:53